Viet Q. Ngo (Ngô Quang Việt)

AI Researcher
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Research Interests

AI researcher and engineer focused on data-efficient, generalizable models for specialized domains. Current research centers on geometric deep learning and protein language models for mutation-effect prediction in protein-ligand and protein-protein interactions. Experience spans NLP and data science, including language modeling, information retrieval, predictive modeling, segmentation, and model interpretability. Long-term goal: doctoral research connecting ML methodology with biomedical and scientific applications.

Education

- **VinUniversity (VinUni)** — Hanoi, Vietnam
  Master of Science in Computer Science (Research Track) — *Sep 2026 - Present*

- **Hanoi University of Science and Technology** — Hanoi, Vietnam
  Engineer's Degree in Computer Science (Global ICT Program) — *Sep 2019 - Sep 2024*
  Grade: Distinction (CPA 3.46/4.0)
  Thesis: "Open-domain Named Entity Recognition for Low-Resource Languages: A Case Study on Vietnamese"

- **Vingroup AI Engineer Training Program (Batch 3)** — Jul - Nov 2023
  The Institute of Big Data (VinBigData)
  Selected for a competitive, fully funded AI/ML training program; completed intensive coursework in machine learning, deep learning, NLP, computer vision, and mathematical foundations.

Publications

- **Open-domain Named Entity Recognition for Low Resource Languages - A Case Study on Vietnamese**: Viet Q. Ngo, Huong T. Le. In Proceedings of the 38th Pacific Asia Conference on Language, Information and Computation (PACLIC), 2024
  - Two-stage multi-task fine-tuning of mT0-large that lifted Vietnamese open-domain NER F1 to 0.75 (VLSP 2021) and 0.94 (PhoNER-COVID19), beating prior published baselines; released a new 125K-sample dataset covering 3,522 entity types.

Research Experience

- **Geometric Deep Learning for Protein Mutation Analysis** — May 2026 - Present
  Advisor: Prof. Truong-Son Hy (HySonLab, University of Alabama at Birmingham) — *Manuscript in preparation*
  - Investigating sequence-only $\Delta\Delta G$ prediction: distilling 3D structural and interface-surface geometry into sequence-conditioned representations at training time, enabling mutation-effect prediction from sequence alone at inference and avoiding costly mutant 3D structure generation for protein-ligand and protein-protein interactions.

- **The Institute of Big Data (VinBigData)** — Hanoi, Vietnam
  NLP Research Engineer — *Dec 2023 - May 2024*
  - Surveyed recent advances in information retrieval, comparing retrieval architectures and summarizing key design trade-offs across the research literature.
  - Investigated named entity recognition for low-resource languages, with a focus on Bahasa and Vietnamese.

Professional Experience

- **Zeder Vietnam Co., Ltd.** — Hanoi, Vietnam
  AI and Automation Engineer — *Jun 2026 - Sep 2026*
  - Owned internal workflow-automation projects end to end: framed business pain points as technical problems, gathered requirements, planned and scoped the work, and implemented solutions independently.

- **VMO Holdings** — Hanoi, Vietnam
  AI Engineer — *Sep 2024 - May 2026*
  - Designed the architecture and end-to-end flow for LLM-based decision-support systems on domain-specific business data (data preparation, prompt design, integration), delivered as production RESTful APIs with FastAPI.

- Applied data science and machine learning techniques to pharmaceutical commercial analytics: building predictive and segmentation models that replaced heuristic sales targets with data-driven benchmarks and surfaced interpretable engagement segments, then translating model outputs into actionable field recommendations.
- Built self-service BI dashboards for the sales team (Tableau + Python data pipelines): gathered requirements and domain knowledge from business teams, coordinated with data engineering to source the right data, then designed the flow, processed and aggregated the data, and built the visualizations.

- **MindX Technology School** Hanoi, Vietnam
  Lecturer (part-time) *Mar 2021 - Oct 2023*
  - Instructed and mentored 100+ students in Python, Data Structures & Algorithms, OOP, and Data Analysis.

- **Hanoi University of Science and Technology, School of ICT**
  Teaching Assistant *Oct 2022 - Mar 2023*
  - Supported university instruction for Database Systems and Programming Techniques; mentored students with code review and technical problem-solving.

Selected Projects

- **Domain-Specific LLM Adaptation**
  Expanded a LLaMA-70B tokenizer vocabulary by 300K tokens (0.5 compression ratio) and ran continual pre-training on the SambaNova platform for specialized Japanese internal-policy documents, improving inference efficiency for domain-specific content.
- **AI-Driven Evidence Retrieval & Verification**
  Engineered a RAG + LLM-as-reranker pipeline producing calibrated, interpretable triage verdicts with confidence scores from high-authority sources via multi-stage reasoning; built retrieval and reranking with OpenAI embeddings, served via FastAPI.
- **Data-Driven Sales Resource Allocation**
  Replaced heuristic activity targets with data-driven benchmarks: trained LightGBM revenue-response models with Optuna tuning and validation R^2 quality gates, then derived interpretable recommendations from SHAP response curves under feasibility caps.
- **HCP Segmentation for Pharmaceutical Field Engagement**
  Built a modeling-ready table from prescription, patient-potential, call-activity, and network features; applied clustering and feature selection to surface interpretable engagement segments.

Technical Skills

- **Research-Relevant ML**: Geometric Deep Learning, Graph Neural Networks, Protein Language Models
- **LLM & NLP Systems**: LLM Fine-Tuning, Continual Pre-Training, Tokenizer Training, Retrieval-Augmented Generation, Domain Adaptation, Structured Outputs
- **Data Science & Analytics**: Predictive Modeling, Clustering & Segmentation, Model Interpretability (SHAP), Hyperparameter Optimization (Optuna)
- **Libraries & Frameworks**: PyTorch, Hugging Face Transformers, scikit-learn, LangChain, LightGBM, XGBoost, NumPy, Pandas
- **Programming & Tools**: Python, C++, SQL, FastAPI/REST, Git, Docker, Linux, CI/CD

Honors and Additional Information

- Academic Excellence Scholarship (Level A), HUST in 3 consecutive semesters (2022-2023)
- Second Prize, Provincial Mathematics Olympiad (Lao Cai), 2019